A Multimedia Quality-Driven Network Resource Management Architecture for Wireless Sensor Networks With Stream Authentication

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***Abstract*—Media integrity, transmission quality, and energy efficiency are critical for secure wireless image streaming in a wireless multimedia sensor network (WMSN). However, conventional data authentication and resource allocation schemes cannot be applied directly to WMSN due to the constraints on limited energy and computing resources. In this paper, we propose a quality-driven scheme to optimize stream authentication and unequal error protection (UEP) jointly. This scheme can provide digital image authentication, image transmission quality optimization, and high energy efficiency for WMSN. The contribution of this research is two-fold as summarized below. First, a new resource allocation-aware greedy stream authentication approach is proposed to simplify the authentication process. Second, an authentication-aware wireless network resource allocation scheme is developed to reduce image distortion and energy consumption in transmission. The scheme is studied by unequally protected image packets with the consideration of coding and authentication dependency. Simulation results demonstrate that the proposed scheme achieves a performance gain of $3 \sim 5$ dB in terms of authenticated image distortion.**

***Index Terms*—Cross layer resource allocation, stream authentication, wireless multimedia sensor network.**

I. Introduction

SECURE and robust multimedia communications become increasingly important for energy-constrained wireless multimedia sensor networks (WMSNs), where security, transmission quality, and energy efficiency are critical. Due to the constraints on energy and computing resources in sensor nodes, WMSN can only provide limited quality-of-service (QoS) and integrity guarantee. Those constraints pose great challenges to WMSN development and motivate us to design proper authentication strategy, cross layer network protocol, and scheduling algorithm for WMSNs.

Conventional binary data authentication schemes could provide data integrity in a strict sense regardless of multimedia content. However, those schemes are not applicable to WMSNs since simple bit-flips due to transmission errors could change the semantic meaning of the multimedia content [1], [2]. On the other hand, traditional watermarking-based multimedia content authentication schemes are robust against bit-errors, packet-losses, and compression distortion. However, embedding secure bits in the frequency band coefficients creates extra source coding overheads and complicates transmission protocol design based on unequal error protection (UEP) in WMSNs [3]. Stream authentication is an option for the WMSN integrity provision due to the fact described as follows. First, authentication algorithms used at the stream level (i.e., post compression) can be adapted easily to time-varying wireless channels with its advantages over traditional content-level (i.e., prior compression) watermarking. Second, stream authentication works without ambiguity since it attaches a one-way hashing function or signature to each packet, such that each packet can be identified clearly as either an authentic or a faked packet [4]–[6]. Finally, as to be discussed in this paper, stream authentication and UEP techniques can work jointly to provide more efficient resource allocation. The more important packets in image data streams can be given higher quality assurance and greater delivery protection.

The joint consideration of quality-driven multimedia integrity and energy efficiency for a WMSN is an important research area. The research conducted in [1] suggested to design a multimedia stream authentication scheme by exploiting unequal importance of different image/video packets. The quality of authenticated media, rather than the authentication probability, was optimized by allocating authentication resources unequally according to the packet distortion weight. The ideas proposed in [2] utilized the unequal authentication protection (UAP) by unequally allocating source/channel coding rates and authentication bit budgets in the compressed image streams. In that approach, the verification probability of all packets in the media stream was also optimized with the overall authentication bit budget. The work reported in [4] proposed a content-aware stream authentication scheme to optimize the authenticated media quality with the help of JPEG2000 image codec in lossy channels. An acyclic authentication graph was explored to optimize the trade-off between the expected image distortion and the crypto-hash tagging

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cost, through the computation based on packet loss probability and visual importance level of the image packets. The research done in [5] proposed a rate-distortion optimization framework with stream authentication for H.264 video transmissions. A video packet transmission scheduler was designed to minimize the visual distortion under the limitation of total bit budget and authentication dependency. Many other related works on error robust multimedia stream authentication were given in [6]–[9]. However, all of the aforementioned approaches are not able to be applied directly to WMSNs because of the energy constraint, which is critical to WMSNs. Additionally, the resource allocation in those approaches was confined to the allocation of authentication bits at the application layer only, resulting in a limited design space and performance gain. Based on our preliminary research work published in [3], we demonstrated the potential quality gain by adjusting authentication and network transmission parameters for each individual image packet. However, this paper provides a comprehensive analytical study and an improved methodological investigation. Here, we propose a methodology which unifies three WMSN design components of: energy minimization, distortion reduction, and authentication provision. By efficiently classifying multimedia packets into various service classes, the proposed approach not only considers the stream authentication at the application layer, but also implements the network resource allocation at the lower layers to provide unequal protection and to achieve a significant performance gain.

The significance of this research is twofold and can be summarized as follows. First, we propose a greedy authentication scheme suitable for UEP-based network resource allocation. Our approach significantly reduces the authentication complexity and authentication dependency overhead, to provide high-level guarantee for image data integrity. Second, an authentication-aware network resource management scheme is proposed, considering both an inborn image decoding dependency and an extra authentication hash link verification dependency. By allocating network and communication resources unequally for different image packets attached with crypto-hash signatures, the authenticated image distortion is greatly reduced without increasing energy consumption overhead.

The rest of this paper is outlined as follows. Section II formulates the WMSN design problem in particular for the optimal energy-constrained, quality-driven image transmission, and authentication. In Section III, the formulated problem is analyzed. A greedy authentication mechanism is proposed to reduce the image authentication complexity. In Section IV, the lower layer energy consumption in WMSN is modeled and minimized using a methodology proposed for the authentication-aware network resource management. Section V offers a system-level complexity analysis on the proposed stream authentication and resource allocation framework. Simulation results are given in Section VI, followed by the conclusions drawn in Section VII.

II. Quality-Authentication-Energy-Driven Cross Layer Problem Formulation

The major symbols and nomenclatures used throughout this paper are summarized in Table I. In this paper, we use a generic model of non-ambiguous image authentication as introduced in [1] and [2]. In this model, the image is reconstructed exclusively from the packets which are received, decoded, and verified at a receiver. To maximize the expected total distortion reduction (i.e., to maximize the quality or to minimize the total distortion), one can select the crypto-hash link parents and control the wireless network resource management parameters jointly (such as modulation, retransmission, and power level), or

TABLE I
Summary of Major Nomenclature and Symbols

$\varepsilon[\Delta D]$	The total expectation of distortion reduction (quality) of the decoded image after packet-by-packet authentication
E_{tot}	The total energy consumption of transmitting the image with authentication and resource allocation applied
$\Delta D_{l,i}$, $E_{l,i}$	The distortion reduction and delivery energy consumption of the i-th packet in layer l
L, N_l	The total number of layers and the number of image packets in layer l of the compressed code stream
$a_{l,i}$	The hash authentication chain allocation of the i-th packet in layer l
$v\,(.)$	The verification probability for a packet with specific hash authentication chain allocation
$\lvert\pi_{l,i}\rvert$	Crypto-hash tag redundancy degree of the i-th packet in layer l
ς	Packet error rate of one image packet
θ	The cumulative probability of delivering an image packet successfully which also depends on the packet error rate of ancestor packets
$S_{l,i}$	The original payload size of the i-th packet in layer l
S_{sig}, S_{hash}	The sizes of signature tag and crypto-hash tag
T_o, T_{cyc}, H_o, H_a	Link layer packet transmission time overhead, sleep-wakeup scheduling cycle time, fragmentation and acknowledgement overhead
e	Desirable BER expectation of PHY
$m_{\max}$, r	Link layer retry limit and PHY transmission rate
P_t, P_r, P_s	Tx/Rx/Sleep power in PHY
R_s, N_0, b, A	Symbol rate, noise power density, modulation constellation size, and channel state factor including attenuation, fading and antenna gain

$$\left\{a_{l,i}, r_{l,i}, m_{\max_{l,i}}, P_{t_{l,i}}\right\}_{l\in\{0,1,\ldots,L-1\}, i\in\{0,1,\ldots,N_l-1\}} = \arg\max\left\{\varepsilon[\Delta D]\right\} \quad (1)$$

subject to the total energy budget constraint given by

$$E_{tot} \le E_{\max}. \quad (2)$$

Given a packet distortion reduction requirement $\Delta D_{l,i}$, a cumulative packet delivery probability $\theta_{l,i}$, and a verification probability $v\,(a_{l,i})$ with respect to the crypto-hash parent selection $a_{l,i}$, one can express the expected authenticated distortion reduction as follows:

$$\varepsilon[\Delta D] = \sum_{l=0}^{L-1}\sum_{i=0}^{N_l-1} \Delta D_{l,i}\theta_{l,i} v\,(a_{l,i}). \quad (3)$$

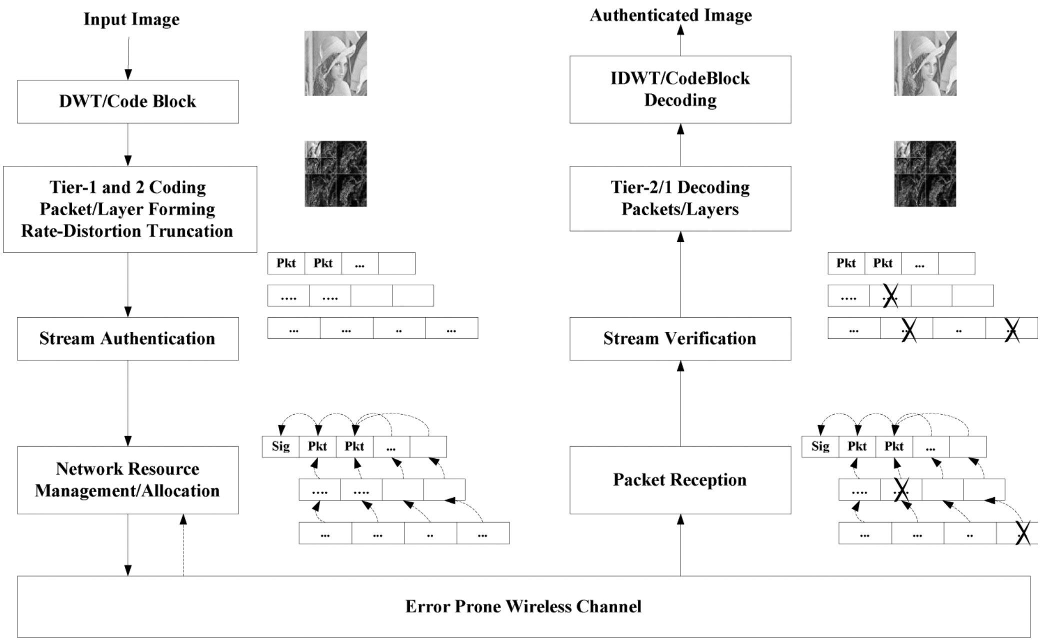

Fig. 1. Quality-driven resource management with stream authentication for digital image transmission.

The evaluation of the image packet distortion reduction can be done based on the image decoding quality gain [10]–[12]. In a JPEG2000 codec, such evaluation can be conducted with the wavelet coefficient square error unit [2], [4], [13]. Furthermore, the communication energy consumption can be evaluated in terms of the sum of link layer energy consumed for delivering each image packet, or

$$E_{tot} = \sum_{l=0}^{L-1} \sum_{i=0}^{N_l-1} E_{l,i}. \tag{4}$$

This problem can be decomposed into several sub-problems and solved by quantitatively analyzing three key parameters: 1) the inborn decoding dependency which is used to find the cumulative packet delivery probability $\theta_{l,i}$, 2) the additional authentication dependency for determining the verification probability $v\left(a_{l,i}\right)$, and 3) the packet delivery energy consumption $E_{l,i}$ at the link layer of WMSN. In Section III, we analyze the first two parameters. We will provide the WMSN link layer energy-quality modeling in Section IV. The overall framework for a quality-driven network resource management system with image stream authentication is illustrated in Fig. 1.

III. Resource Allocation-Aware Authentication Hash Link Allocation

A. Authentication Dependency and Distortion Analysis for JPEG2000 Coding

A large number of image packets are present in a compressed image stream, and therefore it is impractical to design an overall optimal authentication hash graph for all packets in a low-cost WMSN. Furthermore, the image packets are interdependent due to the decoding dependency, which provides a great potential to simplify the stream authentication design. Without losing generality, we carry out our study based on the JPEG2000 [14], [15] image compression standard. In JPEG2000, the decoding of each high-quality-layer packet depends on the successful decoding of those packets in the same code block or precinct from the previous quality layers. Thus, it is reasonable to match the crypto-hash link dependency assignment during the authentication process to the image decoding dependency in a way similar to what was suggested in [4]. Since the crypto-hash tag of the high-quality-layer's packet is attached to the previous layer's packet, the authentication rate-distortion performance can be optimized and there is no extra authentication dependency overhead incurred.

Through this nontrivial unification between image authentication and decoding, the stream authentication process is simplified to the crypto-hash link assignment on Layer 0 packets. Furthermore, only one outgoing crypto-hash link needs to be applied to each packet if the whole image is signed only once. For example, let Ω_i denote the set consisting of the entire authentication ancestors of the ith packet in Layer 0 and the said packet itself. If the crypto-hash tag of Packet 4 $(i=4)$ is attached to Packet 2 $(i=2)$ and the tag of Packet 2 is attached to Packet 0 ($i=0$, i.e., the root signature packet), then we have the ancestor set $\Omega_4=\{0,2,4\}$. The authenticated image quality is shown as

$$\begin{aligned}\varepsilon[\Delta D] =\ & \Delta D_{0,0} \times (1-\varsigma_{0,0}) \\ & + \sum_{i=1}^{N_0-1} \Delta D_{0,i} \prod_{j\in\Omega_i} \left(1-\varsigma_{0,j}\right) \\ & + \sum_{l=1}^{L-1} \sum_{i=0}^{N_l-1} \Delta D_{l,i} \prod_{k=0}^{l} \left(1-\varsigma_{k,i}\right) \\ & \times \prod_{j\in\Omega_i} \left(1-\varsigma_{0,j}\right). \end{aligned} \tag{5}$$

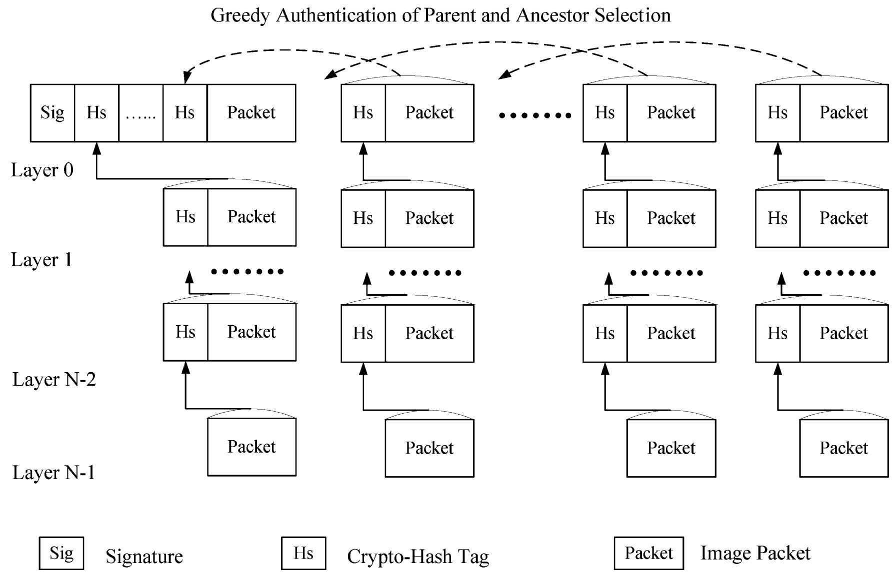

Fig. 2. Illustration of stream authentication and greedy authentication of Layer 0 packets.

The crypto-hash link assignment for stream authentication generates extra data load in terms of the source encoding overhead (due to the hash tag size S_{hash} and signature size S_{sig}), resulting in higher communication energy consumption, while the extra data dependency incurs less error resilience during transmissions. Let $|\pi_{l,i}|$ denote the incoming hash link redundancy of the ith packet in Layer l. For example, if Packet 2 in Layer 0 has two authentication descendants (i.e., Packet 3 in Layer 0 and Packet 4 in Layer 0), then the redundancy degree of Packet 2 is $|\pi_{0,2}| = |1+1| = 2$. Based on the above analysis, the total energy consumption can be expressed as

$$\begin{aligned} E_{tot} = & E_{0,0}\Big(S_{sig} + |\pi_{0,0}|\, S_{hash} + S_{0,0}\Big) \\ & + \sum_{i=1}^{N_0-1} E_{0,i}\Big(|\pi_{0,i}|\, S_{hash} + S_{0,i}\Big) \\ & + \sum_{l=1}^{L-2}\sum_{i=0}^{N_l-1} E_{l,i}\Big(S_{hash} + S_{l,i}\Big) \\ & + \sum_{i=0}^{N_L-1} E_{L-1,i}\,(S_{L-1,i}). \end{aligned} \tag{6}$$

B. Simple Greedy Authentication

Due to the fact that there are multiple independently encoded packets in each quality layer of the JPEG2000 image stream, it is computationally intensive to build a global optimal authentication graph, and thus is cost-prohibitive for a computing resource constrained sensor node in WMSNs. With the network resource allocation applied to the authenticated image stream, we propose a greedy authentication scheme that significantly reduces the computational complexity and has only a minimal authentication dependency overhead. The proposed greedy authentication scheme is illustrated in Fig. 2.

In the greedy authentication scheme, the crypto-hash tag of each Layer 0 packet is attached to a parent packet to maximize the verification probability (i.e., $\max\{v(a)\}$). The crypto-hash assignment for the ith packet in Layer 0 $a_{0,i}$ is a recursive process, which finds ancestor packets of the parent. Let the final result of the greedy authentication $a_{0,i}$ for the ith packet in Layer 0 be Ω_i. From Fig. 2, it is seen that the selection of the root signature packet as the authentication parent leads to maximal verification probability for the packet itself. The corresponding overhead is to decrease the verification probability of all other Layer 0 packets due to the extra bit rate found in the attached hash tags. For example, if the fourth packet in Layer 0 selects the root signature packet as its parent, then we have $\Omega_4 = \{0, 4\}$. Let the channel BER be e and the payload size of the ith image packet be S_i. Then, the verification probability of the fourth packet can be expressed as

$$v(a_{0,4})\Big|_{greedy} = (1-e)^{S_{sig}+S_{hash}+S_0} \times (1-e)^{S_{hash}+S_4}. \tag{7}$$

It is noted that the second part of this equation, $(1-e)^{S_{hash}+S_4}$, includes the additional bit overhead S_{hash} due to the incoming crypto-hash link for high layer packets. If the fourth packet selects any other packet as its authentication parent, such as Packet 2, the best case (when $\Omega_2 = \{0, 2\}$) verification probability of $v(a_{0,4})$ will still be lower than the above value, or

$$\begin{aligned} v(a_{0,4})\Big|_{nongreedy} = & (1-e)^{S_{sig}+S_{hash}+S_0} \\ & \times (1-e)^{S_{hash}+S_2} \times (1-e)^{S_{hash}+S_4}. \end{aligned} \tag{8}$$

Therefore, the proposed greedy authentication scheme does achieve the best verification probability for each packet. It is noted that this greedy scheme is very simple and has an extremely low complexity. However, the disadvantage of the greedy authentication is increased authentication bit overhead for signature packet, which may negatively affect the verification probability for the whole image stream. Nevertheless, this disadvantage can be compensated by network resource allocation, on which we will discuss in the section followed.

For analysis simplicity, the root signature packet deserves for the best transmission protection since the whole image stream would not be verifiable without this signature. With respect to all other Layer 0 packets, assigning the outgoing hash link to the signature packet in this greedy scheme only incurs minimal authentication dependency. In the next section, we will discuss this topic in detail and propose the methodology for an energy-constrained network resource allocation scheme working jointly with the stream authentication scheme.

IV. Authentication-Aware Energy-Quality-Driven Resource Allocation

The goal of authentication-aware resource management design is to provide the best effort authenticated image transmission quality with a given energy constraint. It includes two objectives, i.e., link layer distortion-energy modeling for authenticated image transmission and the UEP-based resource allocation.

A. Link Layer Distortion-Energy Modeling

In this sub-section, we quantify the relationship between three factors, i.e., the network resource allocation, the image distortion during transmission, and the corresponding communication energy consumption. Here, the network resource allocation scheme includes the PHY layer transmission power P_t, modulation constellation size b, and MAC retry limit $m_{\max}$. The image distortion is affected dominantly by the average packet loss ratio $\bar{\xi}$. Let the average energy consumed by the corresponding image transmission be $\bar{E}$. Such a relationship can be derived as follows. The SNR fluctuates in wireless channels due to multi-path propagation, channel attenuation/fading, and interference. There is a performance tradeoff between BER, transmission rate, and the transmission power. By applying multi-rate-multi-power [16], [17] transmission control, the BER is quantitatively expressed as follows [18]–[20]:

$$e = \frac{2}{b}\left(1 - \frac{1}{2^{b/2}}\right) erfc\left(\sqrt{\frac{3}{2(2^b-1)}\frac{P_t A}{R_s N_0}}\right). \quad (9)$$

In this paper, we consider QPSK and QAM modulation schemes with an even constellation size ($b \in \{2,4,6,8\}$) in a slow fading channel environment. The loss ratio for a packet with length S, given incoming authentication redundancy $|\pi|$ and transmission overhead H_o and H_a, is expressed as

$$\xi = 1 - (1-e)^{S+|\pi|S_{hash}+H_o+H_a}. \quad (10)$$

With the increase of incoming authentication redundancy, both packet loss ratio overhead and energy consumption overhead will also be incremented due to the attached crypto-hash tags. The energy consumption of delivering a packet of length S can be estimated by

$$\begin{aligned} E = {} & P_t \times \left(\frac{S+|\pi|\, S_{hash}+H_o}{R} + T_o\right) \\ & + P_r \times \left(\frac{H_a}{R} + T_o\right) + P_t \times \left(\frac{H_a}{R} + T_o\right) \\ & + P_r \times \left(\frac{S+|\pi|\, S_{hash}+H_o}{R} + T_o\right) \\ & + 2P_s \times \left(T_{cyc} - \frac{S+|\pi|\, S_{hash}+H_o+H_a}{R} - 2T_o\right). \end{aligned} \quad (11)$$

With the link layer retransmission factor, the average packet loss ratio expectation can be summarized using the following expression [21], [22]:

$$\bar{\xi} = \xi^{m_{\max}}. \quad (12)$$

Therefore, the energy consumption expectation [22], [23] can be written as

$$\bar{E} = E \times \frac{1-\xi^{m_{\max}}}{1-\xi}. \quad (13)$$

Thus, the packet delivery distortion-energy performance can be modeled quantitatively as functions $\bar{\xi}$ and $\bar{E}$. These functions should be fine-tuned using transmission control parameters P_t, b, and $m_{\max}$.

B. UEP-Based Authentication-Aware Resource Allocation

The problem for the authenticated image quality maximization can be resolved through adjusting the network transmission parameters for each image packet in respect to each image quality layer of JPEG2000. However, the exhaustive search for an optimal solution is time/resource-consuming in terms of available computing power in sensor nodes, and thus is not suitable for secure image streaming in WMSN. To overcome this dilemma, we design a practical evolutionary algorithm with a low complexity by taking advantage of the common features in network resource allocation and image stream authentication scheme. Specifically, we propose a network resource allocation scheme where the packet dependency in the stream authentication mechanism is fully exploited. The algorithm can be described below. At the beginning, an efficient packet classification is performed according to the QoS requirements for those packets. Based on the greedy authentication mechanism proposed in the previous section, the packets in the image stream are divided into three QoS classes: 1) signature packets, 2) Layer 0 packets without a signature, and 3) high image quality layer packets. Since packets in the same QoS class have a similar contribution in terms of verification and decoding, we assign the same network transmission parameters to the packets belonging to the same QoS class. In this way, the dimension of the solution space and optimization complexity is significantly reduced. On the other hand, while the transmission power control is important to fine-tune the distortion-energy performance, the transmission rate and retransmission limits can only be controlled coarsely [24]. Therefore, we optimize the transmission data rate independently in a way very similar to what used in our previously published work [24]. Therefore, this evolutionary algorithm is proposed and described in details in Algorithm 1. It is noted that the evolutional nature of this algorithm is due to the inherent dependency relationship in the proposed greedy stream authentication structure.

Algorithm 1: Energy-Constrained Resource Allocation for Greedy Image Stream Authentication

1) Define algorithm I/O.
   Input parameters: input image, the number of layers L in the image stream, the number of packets N_l in each layer, the channel state factor A, the rate (or size) $S_{l,i}$, and distortion reduction $\Delta D_{l,i}$ of each image packet, where $l \in \{0, 1, \ldots, L-1\}$, and $i \in \{0, 1, \ldots, N_l - 1\}$.
   Output parameters: Optimized transmission power triplet $\{P_{t(0,0)}, P_{t(0,1)}, P_{t(1,1)}\}$ of the three packet QoS classes.
2) Perform greedy stream authentication against the image stream. Assign signature or hash tag to each packet except the last layer packet. Calculate the incoming redundancy degree $|\pi_{l,i}|$ for each packet. Let ":=" denote the assignment operation and "=" denote the equality judgment.
   If $l = 0$ and $i = 0$, then $\chi := 1$; else $\chi := 0$.
   Update the size of each image packet. $S_{l,i} = S_{l,i} + |\pi_{l,i}| \times S_{hash} + \chi \times S_{sig}$.
3) Classify packets into three QoS categories, translate $\{P_{t(0,0)}, P_{t(0,1)}, P_{t(1,1)}\}$ to desirable BER expectation $\{e_{(0,0)}, e_{(0,1)}, e_{(1,1)}\}$ according to (9). Associate $\{e_{(0,0)}, e_{(0,1)}, e_{(1,1)}\}$ to the three packet QoS classes and perform chromosome coding. Initialize the population with size K for the first iteration. Initialize the iteration indicator $g := 0$ and the maximum iteration limit $G_{\max}$.
   For $k = 0$ to K do: $\{e_{(0,0)}(k), e_{(0,1)}(k), e_{(1,1)}(k)\} := rand(0,1)$.
4) Start the evolution loops to increase the fitness values of chromosomes.
   If $g < G_{\max}$, then:
   Perform iteration of Steps 5–7; $g := g + 1$;
   Else: Go to Step 8.
5) For each element $\{e_{(0,0)}(k), e_{(0,1)}(k), e_{(1,1)}(k)\}$ in the current population K, evaluate the authenticated distortion reduction expectation $\varepsilon[\Delta D](k)$ and the energy consumption expectation $E_{tot}(k)$ of transmitting the authenticated image according to (3) and (4).
6) Evaluate the fitness value $f(k)$ for each element in the current iteration. Let α denote a very small constant value to slightly relax the energy constraint and $f_{\min}$ denote a very small constant value of the minimum fitness limit.
   If $E_{tot}(k) \le E_{\max} + \alpha$, then: $f(k) := \varepsilon[\Delta D](k)$.
   Else: $f(k) := f_{\min}$.
7) Sort the chromosomes in the current population in descending order according to the value of $f(k)$. Calculate the crossover probability $p(k) = f(k) / \sum_{i=0}^{K-1} f(i)$ of each elite parent in the sorted population. Randomly crossover the chromosomes $\{e_{(0,0)}(k), e_{(0,1)}(k), e_{(1,1)}(k)\}_{current}$ in the current population, and produce a new population $\{e_{(0,0)}(k), e_{(0,1)}(k), e_{(1,1)}(k)\}_{next}$ for the next iteration. Go to Step 4.
8) Algorithm is finished. Output the chromosome $\{e_{(0,0)}, e_{(0,1)}, e_{(1,1)}\}$ in the final population with the highest fitness value f, and translate desirable BER in the chromosome into power control parameters $\{P_{t(0,0)}, P_{t(0,1)}, P_{t(1,1)}\}$.

V. Complexity Analysis

The overall complexity of the optimization algorithm for the unified network resource allocation and stream authentication is significantly reduced due to the following two facts. First, the greedy authentication scheme reduces the authentication complexity at the application layer. Second, the proposed QoS-based packet classification reduces the resource allocation complexity at the lower layers.

At the application layer, a post-compression stream authentication is applied to the image packets at different image quality layers of the JPEG2000 code stream. As shown in Section III, the stream authentication problem is actually equivalent to the crypto-hash link assignment problem for Layer 0 packets. Consider n packets in Layer 0 of the coded stream, with each packet selecting an authentication parent in the direction of its root within the authentication graph. It is noted that the authentication graph must be created as an acyclic graph so that a realistic dependency tree is allowed to exist. Without some nontrivial simplification, based on the greedy algorithms proposed in Section III, a straightforward yet naive stream authentication algorithm would have to exhaustively search for all of possible crypto-hash link parents for each packet. For instance, there would be $i-1$ possible crypto-hash parent selection options for the ith packet in Layer 0. Overall, there would be $\sum_{i=1}^{n-1}(i-1)$ comparison operations for the entire stream authentication optimization process. The complexity for a naive optimization process outlined above would therefore be $O(n^2)$. On the other hand, if the greedy algorithm is utilized for stream authentication, as detailed in Section III, each packet in Layer 0 only needs to choose a pre-determined crypto-hash parent. This selection is pre-deterministic since the authentication parent is statistically assigned as the root signature packet. Thus, our proposed greedy authentication algorithm has an extremely low computational complexity $O(1)$ for each packet, and the total complexity is only $O(n)$ for all the n packets to pinpoint the authentication tag.

On the MAC and PHY layers, the network resource allocation is applied to the image coded stream during each transmission. The transmission strategy is adjusted for an optimal energy-distortion performance. It is noted that there are a large number of packets in the image coded stream. As explained below, those packets would incur an unrealistically high overhead if the optimal transmission parameters are calculated comprehensively for every single packets. Consider a typical example of an image coded stream, which consists of three image quality layers with three wavelet frequency sub-bands, as well as five JPEG2000 packets for each image quality layer (i.e., one image packet data carries the low frequency band information, one packet carries the middle frequency band, and three packets carry the three spatial precincts in the high frequency band). The resource allocation algorithm needs to identify the strategies for the 15 packets independently, which is impractical in real-time secure multimedia streaming for a low cost wireless sensor. We hereby suppose that the image data packets are mapped on one-on-one basis to the transmission packets in WMSN, but this complexity analysis is also extendable to a generic case in WMSNs with m packets for an image coded stream. Since the resource allocation optimization algorithm needs to identify the optimal transmission strategies for all packets exhaustively, the total dimension would be m to discover the right transmission parameters. To achieve this, it would be impractical to perform real-time secure multimedia streaming in low-cost wireless sensors. With the proposed packet classification described in Section IV, the image coded stream is divided into three categories. Overall, the network resource allocation is performed on a classification basis, and only three parameters representing the transmission strategy need to be optimized. With the help of this significantly simplified but yet effective optimization algorithm, the resource allocation optimization dimension is reduced from m to three.

VI. SIMULATIONS

In this section, we evaluate the energy consumption overhead and authenticated distortion reduction performance gain in a realistic simulation setup for the proposed greedy authentication scheme working jointly with the authentication-aware network resource allocation scheme (i.e., Greedy+UEP). This proposed scheme is compared against the conventional crypto-hash chain-based authentication scheme with equal error protection (EEP) (i.e., Chain+EEP). The chain-based authentication scheme working jointly with the UEP resource allocation (i.e., Chain+UEP) is also evaluated for comparison. The simulation setup and parameter configuration are explained as follows. At the application layer, the well-known "Lena" image with 256×256 pixels is used as an image example in the simulations. A C/C++ implementation of the JPEG2000 image compression codec standard [25] is utilized. In this codec, there are three image quality layers embedded in the coded stream, and different compression ratios (30, 35, and 40) are applied during each simulation. The default progression order is LRCP (i.e., Layer, Resolution level, Component, Precinct in JPEG2000). In the stream authentication scheme, the SHA-1 algorithm is used for crypto-hash tagging with a size of 160 bits. A digital signature scheme is applied to sign the root packet with 1024 bits of signature overhead. The parameters of network communications are listed as follows. The symbol rate is 1000 kHz and the default channel state factor is -130 dB. The MAC layer fragmentation overhead is six bytes and the acknowledgement overhead is eight bytes. The sleep-wakeup interval is 0.1 s. Control packets are transmitted at 20 mW and the receive power is 15 mW. The sleep power is 10 μW.

In comparison with the chain-based authentication and the EEP resource allocation scheme, Fig. 3 demonstrates the authenticated image distortion reduction performance gain for the proposed scheme of the greedy authentication and UEP network resource allocation. It is observed that the proposed scheme achieves a comfortable performance gain with various energy consumption budgets. The authenticated image quality is incremented when the energy budget increases for each approach. Remarkably, the performance gain of the proposed scheme is especially high under low energy budgets. In those scenarios under strict communication energy budgets, the proposed UEP scheme can allocate communication resources more efficiently to important packets (such as the root signature packet and other Layer 0 packets) than the EEP scheme.

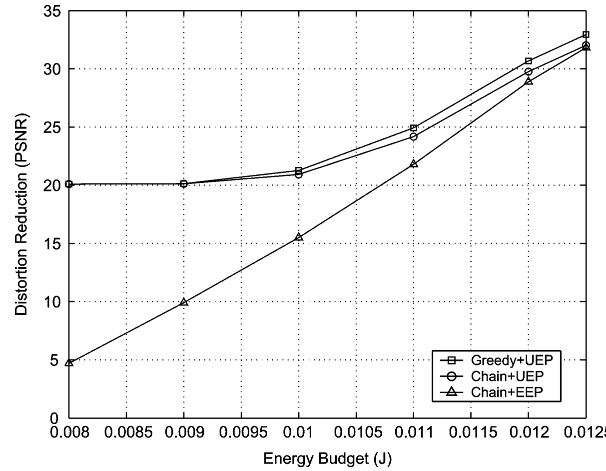

Fig. 3. Authenticated image quality and energy budget for compression ratio 30:1.

The causes for authentication-distortion-energy performance improvement are summarized as follows. First, the proposed greedy stream authentication scheme is much more robust against channel errors because its inter-packet dependency overhead is much lower than the overhead of the "Chain+EEP" scheme. In the "Chain+EEP" scheme, decoding of all image packets depends on the successful decoding of both root signature packet and all preceding authentication packets. In contrast, packet decoding in the greedy authentication scheme depends only on the root signature packet. Although the bit rate overhead for the root packet is increased due to the additionally attached hash tags, which are needed by other Layer 0 packets in the greedy authentication scheme, the inter-packet dependency overhead is reduced immensely. Second, the proposed UEP-based network resource allocation scheme efficiently protects the image stream against transmission errors, where both image decoding dependency and authentication dependency are exploited. Specifically, the root signature packet and other Layer 0 packets are protected with more communication energy in order to improve the authenticated image quality, while other less important packets are given fewer resources to save communication energy.

In terms of actual energy consumptions, Fig. 4 compares the performance of the proposed "Greedy+UEP" scheme against the "Chain+EEP" scheme and the "Chain+UEP" scheme. Furthermore, Figs. 5 and 6 demonstrate the energy consumption and authenticated distortion reduction performances under various image compression ratios in different scenarios. In those figures, it is observed that the UEP-based resource allocation scheme is very effective to achieve a great authentication-distortion-energy performance gain, particularly when it works jointly with a simple greedy stream authentication scheme. In this scheme, the packet distortion importance, inter-packet decoding dependency, and authentication dependency are exploited successfully and efficiently. In contrast, EEP transmission scheme results in a low energy efficiency due to the waste of excessive communication resources for those important packets in the authentication and decoding process. On the other hand, even though the chain-based authentication

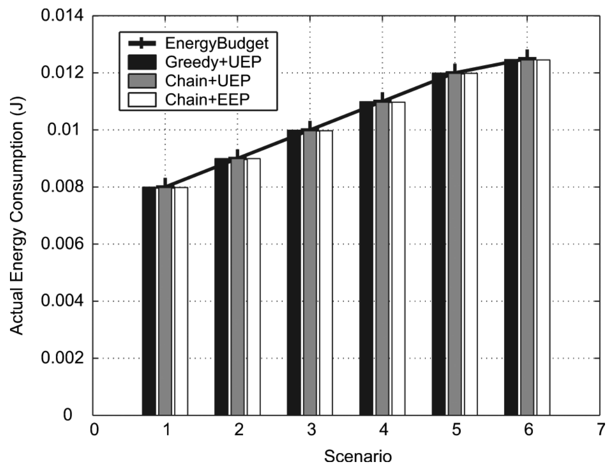

Fig. 4. Actual energy consumption and energy budget for compression ratio 30:1.

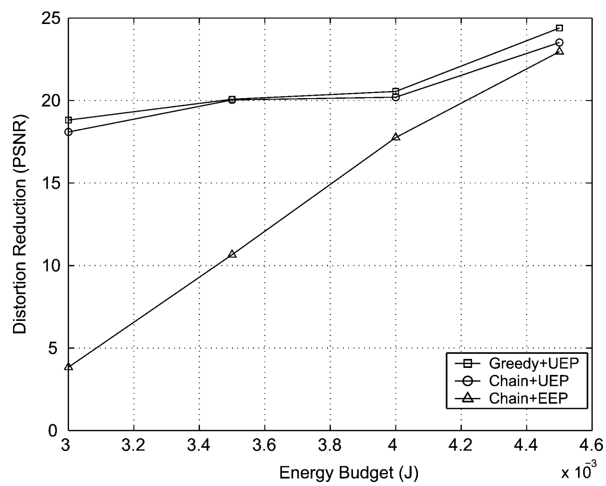

Fig. 5. Authenticated image quality and energy budget for compression ratio 35:1.

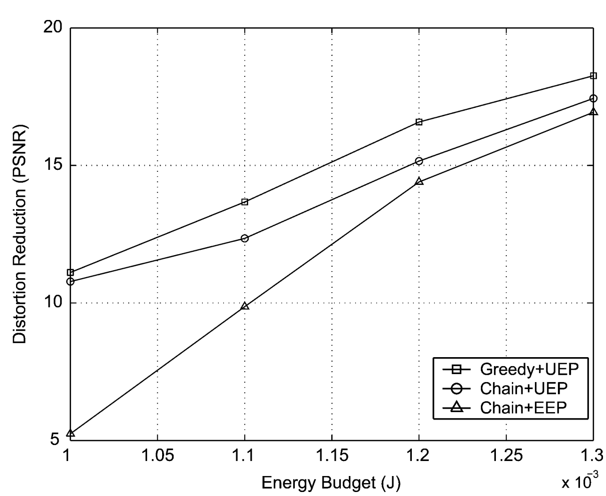

Fig. 6. Authenticated image quality and energy budget for compression ratio 40:1.

in the "Chain+UEP" scheme introduces more overhead for authentication dependency, the UEP-based transmission at lower layers can still compensate the losses in robustness.

VII. Conclusion

In this paper, we have proposed a methodology for quality-driven and energy-efficient transmission of authenticated images in WMSNs. First, a JPEG2000 compatible stream authentication scheme is proposed with a minimal authentication dependency overhead, which is very easy to be integrated with network resource allocation schemes in order to tackle the problem of severe energy constraints in WMSNs. Furthermore, a general UEP-based network resource allocation framework is developed to optimize the image transmission quality with integrity and energy efficiency assurance. Our simulation results demonstrated that the proposed schemes significantly improved the authenticated image quality even under strict communication energy consumption constraints in wireless multimedia sensor networks.

References

[1] Q. Sun, J. Apostolopoulos, C.-W. Chen, and S.-F. Chang, "Quality-optimized and secure end-to-end authentication for media delivery," *Proc. IEEE*, vol. 96, no. 1, pp. 97–110, Jan. 2008.

[2] Z. Li, Q. Sun, Y. Lian, and C.-W. Chen, "Joint source-channel-authentication resource allocation and unequal authenticity protection for multimedia over wireless networks," *IEEE Trans. Multimedia*, vol. 9, no. 4, pp. 837–850, Jun. 2007.

[3] W. Wang, D. Peng, H. Wang, H. Sharif, and H.-H. Chen, "Energy-distortion-authentication optimized resource allocation for secure wireless image streaming," in *Proc. IEEE WCNC*, Apr. 2008, pp. 2810–2815.

[4] Z. Zhang, Q. Sun, W. Wong, J. Apostolopoulos, and S. Wee, "An optimized content-aware authentication scheme for streaming JPEG-2000 images over lossy networks," *IEEE Trans. Multimedia*, vol. 9, no. 2, pp. 320–331, Feb. 2007.

[5] Z. Zhang, Q. Sun, W. Wong, J. Apostolopoulos, and S. Wee, "Rate-distortion-authentication optimized streaming of authenticated video," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 17, no. 5, pp. 544–557, May 2007.

[6] Q. Sun and S.-F. Chang, "A secure and robust digital signature scheme for JPEG2000 image authentication," *IEEE Trans. Multimedia*, vol. 7, no. 3, pp. 480–494, Jun. 2005.

[7] Q. Sun, D. He, and Q. Tian, "A secure and robust authentication scheme for video transcoding," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 16, no. 10, pp. 1232–1244, Oct. 2006.

[8] C.-K. Wong and S. Lam, "Digital signatures for flows and multicasts," *IEEE/ACM Trans. Netw.*, vol. 7, no. 4, pp. 502–513, Aug. 1999.

[9] D. Skraparlis, "Design of an efficient authentication method for modern image and video," *IEEE Trans. Consum. Electron.*, vol. 49, no. 2, pp. 417–426, May 2003.

[10] M. van der Schaar, D. Turaga, and R. Wong, "Classification-based system for cross-layer optimized wireless video transmission," *IEEE Trans. Multimedia*, vol. 8, no. 5, pp. 1082–1095, Oct. 2006.

[11] P. Chou and Z. Miao, "Rate-distortion optimized streaming of packetized media," *IEEE Trans. Multimedia.*, vol. 8, no. 2, pp. 390–404, Apr. 2006.

[12] T. Ozcelebi, A. Tekalp, and M. Civanlar, "Delay-distortion optimization for content-adaptive video streaming," *IEEE Trans. Multimedia*, vol. 9, no. 4, pp. 826–836, Jun. 2007.

[13] D. Taubman, "High performance scalable image compression with EBCOT," *IEEE Trans. Image Process.*, vol. 9, no. 7, pp. 1158–1170, Jul. 2000.

[14] K. Varma and A. Bell, "JPEG2000—choices and tradeoffs for encoders," *IEEE Signal Process. Mag.*, vol. 21, no. 6, pp. 70–75, Nov. 2004.

[15] C. Christopoulos, A. Skodras, and T. Ebrahimi, "The JPEG2000 still image coding system: An overview," *IEEE Trans. Consum. Electron.*, vol. 46, no. 4, pp. 1103–1127, Nov. 2000.

[16] D. Qiao, S. Choi, A. Jain, and K. Shin, "Miser: An optimal low-energy transmission strategy for IEEE 802.11a/h," in *Proc. ACM MobiCom*, Sep. 2003, pp. 161–175.

[17] D. Qiao and K. Shin, "Energy-efficient airtime allocation in multi-rate multi-power-level wireless lans," in *Proc. ACM QShine*, Aug. 2007.

[18] C. Schurgers, O. Aberthorne, and M. Srivastava, "Modulation scaling for energy aware communication systems," in *Proc. Low Power Electronics and Design*, Aug. 2001, pp. 96–99.

[19] H. Simon*, Communication System*, 3rd ed. New York: Wiley, 1994, pp. 510–553.

[20] S. William*, Data and Computer Communications*, 7th ed. Upper Saddle River, NJ: Prentice-Hall, 2000, pp. 85–86.

[21] L. Qiong and M. van der Schaar, "Providing adaptive QoS to layered video over wireless local area networks through real-time retry limit adaptation," *IEEE Trans. Multimedia*, vol. 6, no. 2, pp. 278–290, Apr. 2004.

[22] M. van der Schaar, D. Turaga, and R. Wong, "Classification-based system for cross-layer optimized wireless video transmission," *IEEE Trans. Multimedia*, vol. 8, no. 5, pp. 1082–1095, Oct. 2006.

[23] M. van der Schaar and D. Turaga, "Cross-layer packetization and retransmission strategies for delay-sensitive wireless multimedia transmission," *IEEE Trans. Multimedia*, vol. 9, no. 1, pp. 185–197, Jan. 2007.

[24] W. Wang, D. Peng, H. Wang, H. Sharif, and H.-H. Chen, "Energy constrained distortion reduction optimization for wavelet-based coded image transmission in wireless sensor networks," *IEEE Trans. Multimedia*, vol. 10, no. 6, pp. 1169–1180, Oct. 2008.

[25] [Online]. Available: http://www.openjpeg.org/.

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